GSA Annual Meeting in Seattle, Washington, USA - 2017

Paper No. 392-25
Presentation Time: 9:00 AM-6:30 PM

3D SCANNING AND PRINTING OF VOLCANIC MONITORING DEVICES TO EVALUATE POTENTIAL ALTERNATIVE CASING

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Volcanic sulfur, gases, and thermal heat can be disruptive and damaging to critical infrastructure that supports monitoring stations on and around an active volcano. In addition, as stations are exposed to the effects of volcanic activity, the environment suffers from the contamination of materials being eroded away. As live monitoring stations become established, it is evident that a range of mitigation strategies are needed to reduce the harmful effects on the equipment and environment.

High resolution images were taken of TinyTag data loggers, using 3D scanners such as Kinect, MakerBot Digitizer, and Next Engine before and after their deployment into the field of an active volcano located at the Hellenic Volcanic Arc on the island of Nisyros, South Aegean. The images revealed that after six months of being submerged in the soil around the caldera, the cases were negatively affected and had corroded, making data retrieval challenging. The original weight of the data logger was approximately 3.92oz where the final weight of the data logger was 3.64oz, a total of 7.14% volume loss. Various alternative materials such as ASA, PEX, Carbon PLA, EPDM, ABS, ASA, and TPE were tested in the lab as a potential solution to protecting the integrity of the data loggers. Several materials such as Carbon PLA, TPE, ASA, and TPU 3D printed cases were made using Prusa i3 printer and a Lulzbot Taz 6 printer. These cases were then deployed at either the Nisyros Caldera or in lab mock environments to see if the alternative materials would survive the harsh environments.

Session No. 392--Booth# 490

D37. Volcanology (Posters)
Wednesday, 25 October 2017: 9:00 AM-6:30 PM

Halls 4EF (Washington State Convention Center)

Geological Society of America *Abstracts with Programs*. Vol. 49, No. 6
doi: 10.1130/abs/2017AM-304844